

RAG Evaluation & Production Readiness

How to Measure, Improve, and Deploy Reliable RAG Systems

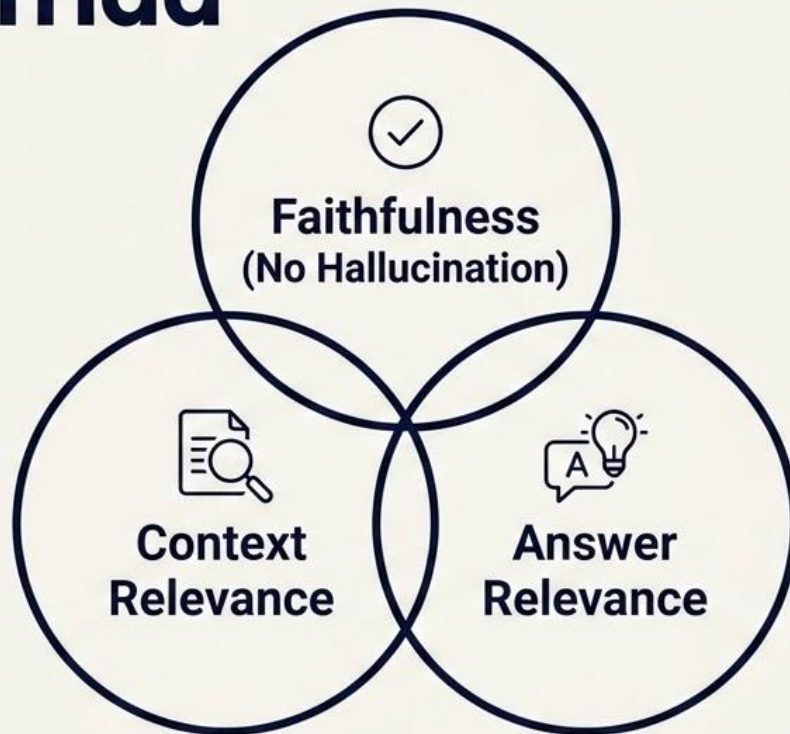


Why RAG Evaluation is Challenging

- Subjective answers
- No single ground truth
- Cost of human evaluation
- Fast iteration cycles



The RAG Triad



1. Faithfulness

- **Definition:** Answer must be grounded in retrieved context
- **How to measure:** LLM-as-Judge
- **Common Issue:** Hallucinations



HALLUCINATED ANSWER
(Before)



GROUNDING ANSWER
(After)

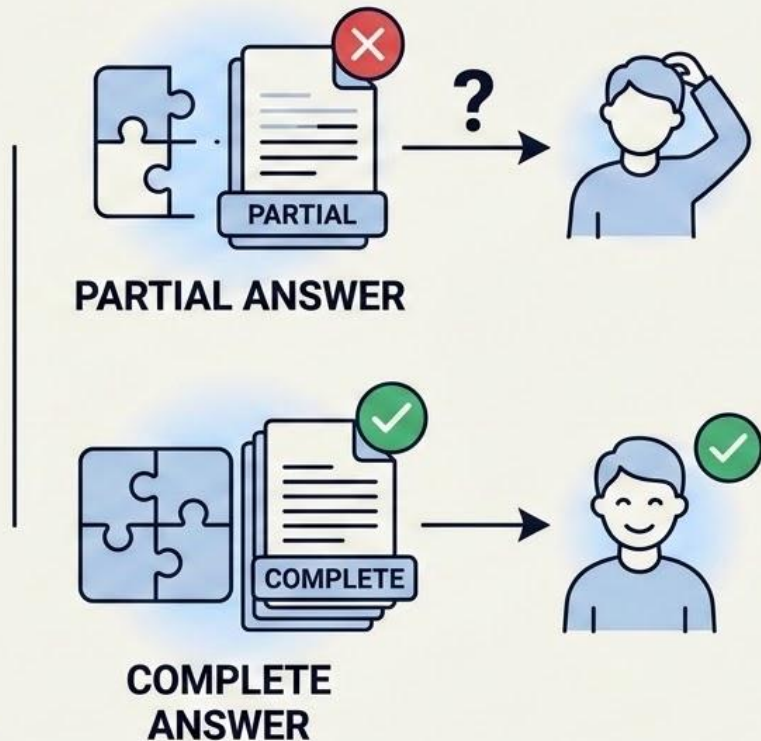
2. Context Relevance

- **Definition:** Retrieved chunks must be relevant to the query
- **Why it matters:**
Garbage in =
Garbage out



3. Answer Relevance

- **Definition:** Final answer must address the user's query properly
- **Focus:** Usefulness and completeness.



Deterministic Retrieval Metrics

Precision@k, Recall@k, F1

Measures relevant content in top **k** results

Hit Rate@k

Percentage of queries with **any relevant result in top k**

MRR (Mean Reciprocal Rank)

Reciprocal of the rank of the first relevant result

NDCG@k (Normalized Discounted Cumulative Gain)

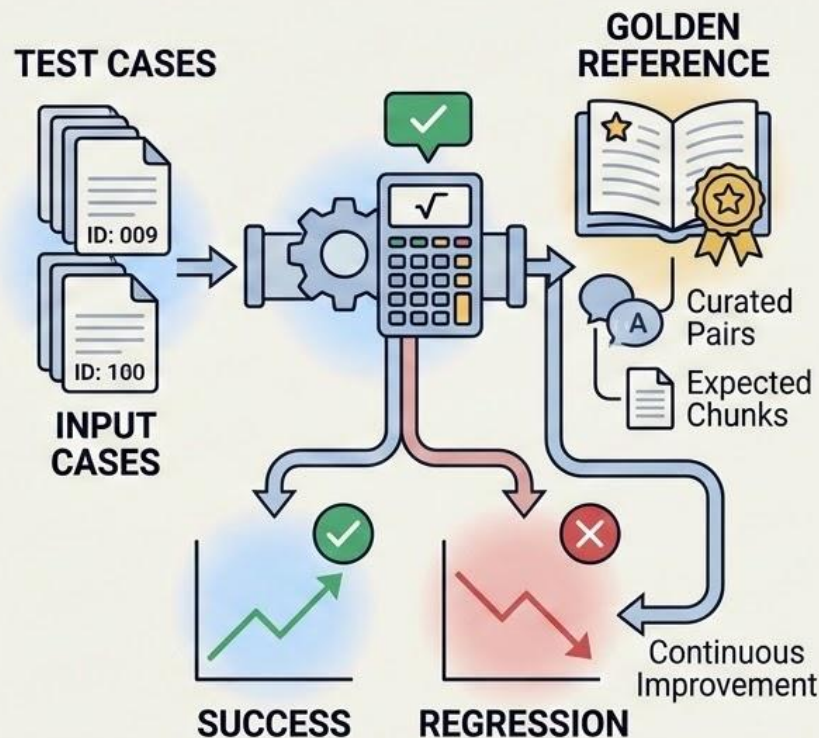
Quality and rank of **all results**

Metric	Focus
P/R/F1	Accuracy & Coverage
Hit Rate	Found/Not Found
MRR	Rank position of 1st hit
NDCG	Overall Rank & Utility



Golden Datasets

- **Importance:**
- Q&A pairs with expected chunks
- Used for regression testing
- Continuous evaluation



Popular RAG Evaluation Tools

- **Ragas** (De-facto standard)
- **LangSmith** (Tracing + Evaluation)
- **Phoenix / Arize**
- **DeepEval**



Production Readiness Checklist



- Automated evaluation pipeline



- Monitoring & observability



- Cost & latency tracking



- Version control & rollback



- A/B testing

Best Practices for Production RAG

- Always use Golden Dataset
- Run evaluation on every change
- Monitor in production
- Set up alerts
- Iterate based on metrics

Key Takeaways

- Evaluation is critical for reliable RAG
- Use RAG Triad + Ragas for LLM-as-Judge
- Build Golden Datasets for regression testing
- Production RAG needs monitoring + continuous improvement

“Great RAG is Measured, Not Assumed” 